Vector spaces, Banach and Hilbert spaces

An overview

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Course: Operator Theory

Presentation Overview

1. Vector Spaces
2. Normed Vector Space
3. Metric Space
4. Hilbert spaces
5. Operators and linear functionals

Definition of Vector Spaces

A *vector space* (or *linear space*) over a field $\mathbb{F}$ is a set V together with two operations that satisfy the following eight axioms. The elements of V are called *vectors*, and the elements of $\mathbb{F}$ are called *scalars*. There are two operations:

- **Vector addition**: An operation $+ : V \times V \to V$ that takes any two vectors $\mathbf{u}, \mathbf{v} \in V$ and assigns to them a third vector $\mathbf{u} + \mathbf{v} \in V$.
- **Scalar multiplication**: An operation $\cdot : \mathbb{F} \times V \to V$ that takes any scalar $a \in \mathbb{F}$ and any vector $\mathbf{v} \in V$ and assigns to them a vector $a\mathbf{v} \in V$.

Axioms of the vector space

The vector space V must satisfy the following axioms for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars $a, b \in \mathbb{F}$:

1. **Associativity of addition**: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
2. **Commutativity of addition**: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
3. **Identity element of addition**: There exists an element $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in V$.
4. **Inverse elements of addition**: For each $\mathbf{u} \in V$, there exists an element $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
5. **Compatibility of scalar multiplication with field multiplication**: $a(b\mathbf{u}) = (ab)\mathbf{u}$.

Axioms of the vector space

6. **Identity element of scalar multiplication**: $1\mathbf{u} = \mathbf{u}$, where 1 is the multiplicative identity in $\mathbb{F}$.
7. **Distributivity of scalar multiplication with respect to vector addition**: $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$.
8. **Distributivity of scalar multiplication with respect to field addition**: $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$.

Examples of Vector Spaces

The space of continuous functions on the interval $[a, b]$:

$$C[a, b] = \{f : [a, b] \to \mathbb{R} \mid f \text{ is continuous}\}$$

is a vector space. Some properties:

- Let $f, g \in C[a, b]$. The sum $h(x) = f(x) + g(x)$ is also continuous for all $x \in [a, b]$. Hence,

  $$h \in C[a, b]$$

  This shows that $C[a, b]$ is closed under addition.
- Let $f \in C[a, b]$ and let $c \in \mathbb{R}$ be any scalar. The function defined by $g(x) = c \cdot f(x)$ is also continuous for all $x \in [a, b]$. Thus,

  $$g \in C[a, b]$$

  This verifies that $C[a, b]$ is closed under scalar multiplication.
- Existence of the zero vector: is defined as $z(x) = 0$ for all $x \in [a, b]$, is continuous.

Definition of a Vector Subspace

Let V be a vector space over a field $\mathbb{F}$.
A subset $W \subseteq V$ is called a vector subspace (or simply a *subspace*) of V if W is itself a vector space over $\mathbb{F}$ with the operations of addition and scalar multiplication defined on V.

Criterion to Characterize a Vector Subspace

A non-empty subset $W \subseteq V$ is a subspace of V if and only if it satisfies the following two conditions:

1. **Closed under addition**: For all $\mathbf{u}, \mathbf{v} \in W$, the sum $\mathbf{u} + \mathbf{v} \in W$.
2. **Closed under scalar multiplication**: For all $\mathbf{u} \in W$ and all $a \in \mathbb{F}$, the product $a\mathbf{u} \in W$.

These conditions ensure that W contains the zero vector, as taking $\mathbf{u} = \mathbf{v} = \mathbf{0}$ satisfies the closure under addition, and taking $a = 0$ and any $\mathbf{u} \in W$ satisfies the closure under scalar multiplication.

Examples of Vector Subspaces

- **Trivial subspace**: The set $\{\mathbf{0}\}$ containing only the zero vector.
- **Whole space**: The vector space V itself.
- Let us see a simple example with 2×2 matrices X and consider with zero trace:

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$

$$\boxed{X_0 = \{A \in \mathbb{R}^{2\times 2} : \mathrm{Tr}(A) = 0\}}$$

- (i) Let $A, B \in X_0$. Then $\mathrm{Tr}(A) = a + d = 0$ and $\mathrm{Tr}(B) = e + h = 0$.
- (ii) Therefore, it is true that $\forall A, B \in X_0, \forall \lambda, \mu \in \mathbb{R}$, we have:

$$\lambda A + \mu B = \lambda \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \mu \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} \lambda a + \mu e & \lambda b + \mu f \\ \lambda c + \mu g & \lambda d + \mu h \end{bmatrix}$$

The trace of the element will be:

$$\mathrm{Tr}(\lambda A + \mu B) = (\lambda a + \mu e) + (\lambda d + \mu h) = \lambda \mathrm{Tr}(A) + \mu \mathrm{Tr}(B) = \lambda \cdot 0 + \mu \cdot 0 = 0.$$

Thus, $\lambda A + \mu B \in X_0, \forall \lambda, \mu \in \mathbb{R}$.

Definition of a Normed Vector Space

A *normed vector space* is a vector space V over a field $\mathbb{F}$ (which is typically $\mathbb{R}$ or $\mathbb{C}$) equipped with a function called a *norm*. The norm assigns a non-negative real number to each vector in V, which intuitively represents the length or size of the vector.

A norm on V is a function $\|\cdot\| : V \to \mathbb{R}$ that satisfies the following properties for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars $a \in \mathbb{F}$:

1. **Non-negativity**: $\|\mathbf{u}\| \geq 0$, and $\|\mathbf{u}\| = 0$ if and only if $\mathbf{u} = \mathbf{0}$.
2. **Scalar Multiplication**: $\|a\mathbf{u}\| = |a| \cdot \|\mathbf{u}\|$.
3. **Triangle Inequality**: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

A vector space V equipped with a norm $\|\cdot\|$ is called a *normed vector space* and is denoted by $(V, \|\cdot\|)$.

Examples of Normed Vector Spaces

- **Euclidean space** $\mathbb{R}^n$: The norm of a vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is given by the Euclidean norm
$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$
- **General** p**-norm** on $\mathbb{R}^n$: For $1 \leq p < \infty$, the p-norm of a vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is defined as
$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}.$$
- **Maximum norm** on $\mathbb{R}^n$: The norm of a vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is given by
$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$
- **Space of continuous functions** $C([a, b])$: The norm of a function $f \in C([a, b])$ is given by
$$\|f\|_\infty = \max_{x \in [a,b]} |f(x)|.$$
- L^p **space**: The space of measurable functions f on a domain D such that

$\|f\|_p^p$

$$\|f\|_p = \left(\int_D |f(x)|^p \, dx \right)^{1/p} < \infty.$$

Hölder inequality

Let (p, q) be a couple of conjugate indexes: $\frac{1}{p} + \frac{1}{q} = 1$. Let $f \in L^p$ and $g \in L^q$ then the product $fg \in L^1$ and the following estimate holds

$$\boxed{\|fg\|_1 \leq \|f\|_p \|g\|_q.}$$

A **generalization** of Hölder inequality is given by the following result.

Let $p, q, r \in (0, \infty)$ such that

$$\boxed{\frac{1}{r} = \frac{1}{p} + \frac{1}{q}.}$$

$$\|fg\|_r^r = \int |f(x)\, g(x)|^r$$

If $f \in L^p$ and $g \in L^q$ then $fg \in L^r$ and we have the following estimates

$$\boxed{\|fg\|_r \leq \|f\|_p \|g\|_q.}$$

Definition of a Metric Space

A *metric space* is a set M equipped with a function

$$\boxed{d : M \times M \to \mathbb{R},}$$

called a *metric* or *distance function*, that satisfies the following properties for all $x, y, z \in M$:

1. **Non-negativity**: $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
2. **Symmetry**: $d(x, y) = d(y, x)$.
3. **Triangle Inequality**: $d(x, z) \leq d(x, y) + d(y, z)$.

A set M together with a metric d is called a *metric space* and is denoted by (M, d).

Examples of Metric Spaces

- **Euclidean space** $\mathbb{R}^n$: The standard metric is given by the Euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

  for $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$.
- **Maximum metric** on $\mathbb{R}^n$: The maximum distance is given by

$$d(\mathbf{x}, \mathbf{y}) = \max_{1 \le i \le n} |x_i - y_i|$$

  for $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$.
- **Taxicab metric** on $\mathbb{R}^n$: Also known as the Manhattan distance, it is given by

$$d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} |x_i - y_i|$$

  for $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$.
- **Space of continuous functions** $C([a, b])$: The metric for continuous functions $f, g \in C([a, b])$ can be defined by

$$d(f, g) = \max_{x \in [a,b]} |f(x) - g(x)|.$$

Spherical Neighborhood

Let (M, d) be a metric space, where M is a set and $d : M \times M \to \mathbb{R}$ is a metric on M. A *spherical neighborhood* (or *open ball*) of radius $r > 0$ centered at a point $x \in M$ is the set

$$\boxed{B(x, r) = \{y \in M \mid \overbrace{d(x, y)}^{= \sqrt{|x_i - t_i|}} < r\}.}$$

Open Sets

A subset $\boxed{U \subseteq M \text{ is called } open}$ if for every point $x \in U$, there exists a radius $r > 0$ such that the spherical neighborhood $B(x, r)$ is contained in U:

$$\forall x \in U, \exists r > 0 \text{ such that } \boxed{B(x, r) \subseteq U}.$$

Closed Sets

A subset $C \subseteq M$ is called *closed* if its complement $M \setminus C$ is open.

Equivalently, C is closed if it contains all its limit points, meaning for any sequence $\{x_n\} \subseteq C$ that converges to $x \in M$, we have $x \in C$:

$$\{x_n\} \subseteq C \text{ and } x_n \to x \implies x \in C.$$

Bounded Sets

A subset $A \subseteq M$ is called *bounded* if there exists a point $x \in M$ and a real number $r > 0$ such that A is contained in the spherical neighborhood $B(x, r)$:

$$\exists x \in M, \exists r > 0 \text{ such that } A \subseteq B(x, r).$$

Cauchy Sequences

A sequence $\{x_n\} \subseteq M$ is called a *Cauchy sequence* if for every $\epsilon > 0$, there exists a positive integer N such that for all $m, n \geq N$, the distance $d(x_m, x_n) < \epsilon$:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N, \boxed{d(x_m, x_n) < \epsilon.}$$

Definition of a Banach Space

A *Banach space* is a normed vector space $(V, \|\cdot\|)$ that is complete with respect to the norm $\|\cdot\|$.

This means that every Cauchy sequence in V converges to a limit that is also in V.

Remark

More precisely, a normed vector space $(V, \|\cdot\|)$ is called a Banach space if every Cauchy sequence $\{\mathbf{x}_n\} \subseteq V$ satisfies

$$\text{For every } \epsilon > 0, \text{ there exists } N \in \mathbb{N} \text{ such that for all } m, n \geq N,$$

$$\|\mathbf{x}_m - \mathbf{x}_n\| < \epsilon,$$

and there exists $\mathbf{x} \in V$ such that

$$\lim_{n\to\infty} \mathbf{x}_n = \mathbf{x}.$$

Examples of Banach Spaces

- **Space of continuous functions** $C([a,b])$ with the sup norm

$$\boxed{\|f\|_\infty = \max_{x\in[a,b]} |f(x)|}$$

$$\|f\| = \int_a^b f(x)\,dx$$

  is a Banach space.
- ℓ^p **spaces** for $1 \le p \le \infty$: The set of all sequences $\{x_n\}$ such that

$$\|\{x_n\}\|_p = \left(\sum_{n=1}^{\infty} |x_n|^p\right)^{1/p} < \infty \quad \text{for } 1 \le p < \infty,$$

  and

$$\|\{x_n\}\|_\infty = \sup_n |x_n| < \infty \quad \text{for } p = \infty,$$

  are Banach spaces.
- L^p **spaces** for $1 \le p \le \infty$: The set of all measurable functions f on a domain D such that

$$\|f\|_p = \left(\int_D |f(x)|^p\,dx\right)^{1/p} < \infty \quad \text{for } 1 \le p < \infty,$$

  and

$$\|f\|_\infty = \operatorname{ess\,sup}_{x\in D} |f(x)| < \infty \quad \text{for } p = \infty,$$

  are Banach spaces.

Exercise: Let $0 < p < 1$. Then $\|x\|_p = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}$ is not norm for ℓ^p.

L^p is a Banach space for $p \geq 1$

Solution

To show that $\|-\|_p$ is not a norm for ℓ^p we make an example. We consider

$$x = (1, 0, 0, 0, \ldots), \qquad y = (0, 1, 0, 0, \ldots),$$

$p \in (0,1)$

then $\|x\|_p = \|y\|_p = 1$ but $\|x + y\|_p = 2^{1/p} > 2$, since $1/p > 1$. Therefore

$$\|x + y\|_p > \|x\|_p + \|y\|_p.$$

$\|x\|_p = \left(\sum_{n \geq 1} |x_n|^p\right)^{1/p}$

So the triangle inequality is not satisfied.

$x + y = (1, 1, 0, \ldots 0)$ $\qquad x_1 = x_2 = 1$

$x_1 \quad x_2$

$\|x+y\|_p = \left(|x_1|^p + |x_2|^p\right)^{1/p} = 2^{1/p} \underbrace{|x_1|}_{=1} = 2^{1/p}$

Definition of Vector Spaces with Inner Product (Pre-Hilbert spaces)

A *vector space with an inner product*, also known as an *inner product space*, is a vector space V over a field $\mathbb{F}$ (which is typically $\mathbb{R}$ or $\mathbb{C}$) equipped with an inner product.

Inner product

The inner product is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ that satisfies the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all scalars $a \in \mathbb{F}$:

$\mathbb{R}$

1. **Linearity in the first argument** (Conjugate linearity in the second argument for $\mathbb{C}$):

$$\langle a\mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle.$$

$\langle a u, v \rangle = a \langle u, v \rangle$

For a complex vector space, it is linear in the first argument and conjugate linear in the second:

$$\langle \mathbf{u}, a\mathbf{v} + \mathbf{w} \rangle = \bar{a}\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle,$$

$\langle u, a v \rangle = \bar{a} \langle u, v \rangle$

where $\bar{a}$ denotes the complex conjugate of a.

2. **Symmetry**:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \overline{\langle \mathbf{v}, \mathbf{u} \rangle},$$

where $\overline{\langle \mathbf{v}, \mathbf{u} \rangle}$ denotes the complex conjugate of $\langle \mathbf{v}, \mathbf{u} \rangle$. For real vector spaces, this reduces to

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle.$$

3. **Positive-definiteness**:

$$\langle \mathbf{u}, \mathbf{u} \rangle \geq 0 \quad \text{and} \quad \boxed{\langle \mathbf{u}, \mathbf{u} \rangle = 0 \text{ if and only if } \mathbf{u} = \mathbf{0}.}$$

Examples of Inner Product Spaces

- **Euclidean space** $\mathbb{R}^n$ with the standard inner product:

$$\boxed{\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n,}$$

  where $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$.

- **Complex space** $\mathbb{C}^n$ with the standard inner product:

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 \overline{y_1} + x_2 \overline{y_2} + \cdots + x_n \overline{y_n},$$

  where $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)$.

- **Space of continuous functions** $C([a, b])$ with the inner product:

$$\langle f, g \rangle = \int_a^b f(x) g(x)\, dx.$$

- L^2 **space**: The set of all measurable functions f and g on a domain D such that

$$\boxed{\langle f, g \rangle = \int_D f(x) \overline{g(x)}\, dx}$$

  is finite.

Also possible for ℓ^2

$g = f$

$$\|f\|_{L^2}^2 = \int |f(x)|^2\, dx$$

Definition of Hilbert Space

A *Hilbert space* is a complete inner product space. Specifically, a Hilbert space is a vector space H over a field $\mathbb{F}$ (typically $\mathbb{R}$ or $\mathbb{C}$) equipped with an inner product $\langle \cdot, \cdot \rangle$ that is complete, i.e.:
The space H is complete with respect to the norm $\| \cdot \|$ induced by the inner product:

$$\boxed{\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}.}$$

This means that every Cauchy sequence in H converges to a limit that is also in H.

Properties of Vector Spaces with Inner Product

Let $(V, \langle \cdot, \cdot \rangle)$ be a vector space over a field $\mathbb{F}$ with an inner product. The inner product induces several important properties and structures in the space.

A **Norm Induced by Inner Product**: The inner product induces a norm on V defined by

$$\boxed{\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.}$$

This norm satisfies the properties of a norm, including non-negativity, homogeneity, and the triangle inequality.

B **Cauchy-Schwarz Inequality**: For all vectors $\mathbf{u}, \mathbf{v} \in V$, the inner product satisfies

$$\boxed{|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|.}$$

C **Triangle Inequality**: For all vectors $\mathbf{u}, \mathbf{v} \in V$, the norm satisfies

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

D **Pythagorean Theorem**: For all orthogonal vectors $\mathbf{u}, \mathbf{v} \in V$ (i.e., $\langle \mathbf{u}, \mathbf{v} \rangle = 0$), the norm satisfies

$$\boxed{\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.}$$

E **Orthogonality**: Two vectors $\mathbf{u}, \mathbf{v} \in V$ are said to be orthogonal if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

The set of all vectors orthogonal to a given vector forms a subspace called the *orthogonal complement* of that vector.

$V^{\perp}$

Parallelogram Law

Let $(V, \langle \cdot, \cdot \rangle)$ be a pre-Hilbert space, which is a vector space equipped with an inner product $\langle \cdot, \cdot \rangle$. The *parallelogram law* states that for all vectors $\mathbf{u}, \mathbf{v} \in V$, the following equality holds:

$$\boxed{\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\left(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2\right),}$$

where $\|\cdot\|$ denotes the norm induced by the inner product:

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}.$$

L^2 why?

ℓ^2 Hilbert

Exercise: Let $1 \leq p < \infty$. Prove that ℓ^p is a Hilbert space if and only if $p = 2$.

L^p

Solution

is a Hilbert space

We assume that ℓ^p, with the corresponding inner product, is a Hilbert space. Consider $e_n, e_m \in \ell^p$ with $m \neq n$ defined as follows:

$$e_n := (0, \underset{n-1}{\ldots}, 0, 1, 0, \ldots),$$

$$e_m := (0, \underset{m-1}{\ldots}, 0, 1, 0, \ldots).$$

Since ℓ^p is a Hilbert space, by the parallelogram law we have

$$\|e_n + e_m\|_p^2 + \|e_n - e_m\|_p^2 = 2(\underbrace{\|e_n\|_p^2}_{1} + \underbrace{\|e_m\|_p^2}_{1}).$$

That is

$$2^{2/p} + 2^{2/p} = 2^2.$$

The unique solution of this equation is $\boxed{p = 2}$. Thus we conclued that ℓ^p is a Hilbert space if and only if $p = 2$.

Dual Space of a Banach Space

Let $(X, \|\cdot\|)$ be a Banach space. The *dual space* of X, denoted X^*, is the space of all bounded linear functionals on X. A *bounded linear functional* is a linear map $f : X \to \mathbb{F}$ (where $\mathbb{F}$ is the field over which X is defined, typically $\mathbb{R}$ or $\mathbb{C}$) that is bounded, meaning there exists a constant C such that for all $\mathbf{x} \in X$,

$$|f(\mathbf{x})| \leq C\|\mathbf{x}\|.$$

C positive

The norm on the dual space X^* is defined by

$$\|f\|_{X^*} = \sup_{\|\mathbf{x}\| \leq 1} |f(\mathbf{x})|,$$

where $\|f\|_{X^*}$ is the operator norm of f. The dual space X^* is itself a Banach space with respect to this norm.

Dual Space of L^p Spaces

For $1 \leq p < \infty$, consider $L^p(\Omega)$ with its dual space $L^q(\Omega)$, where $\Omega \subseteq \mathbb{R}^n$ and

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Conjugate index of p

For $g \in L^q(\Omega)$, define a functional $F_g : L^p(\Omega) \to \mathbb{F}$ by

$$F_g(f) = \int_\Omega f(x) g(x)\, dx$$

for $f \in L^p(\Omega)$. This functional is linear. It is also continuous with norm (thanks to Hölder inequality)

$\|F_g\|_{L^1}$

$$|F_g(f)| = \left| \int_\Omega f(x) g(x)\, dx \right| \leq \left(\int_\Omega |f|^p dx \right)^{1/p} \left(\int_\Omega |g|^q dx \right)^{1/q}.$$

This example shows how functionals on L^p spaces can be represented using functions in L^q. This is a general fact and we will see the important representation theorem.

Riesz Representation

Let $1 \leq p < \infty$ and let $L^p(\Omega, \mu)$ denote the Lebesgue space of p-integrable functions defined on Ω with measure μ. The dual space of $L^p(\Omega, \mu)$ is given by $L^q(\Omega, \mu)$, where $\frac{1}{p} + \frac{1}{q} = 1$.

Theorem: For $1 \leq p < \infty$, the dual space of $L^p(\Omega, \mu)$ is isometrically isomorphic to $L^q(\Omega, \mu)$. That is, for each bounded linear functional $L \in (L^p(\Omega, \mu))^*$, there exists a unique function $g \in L^q(\Omega, \mu)$ such that

$$\boxed{L(f) = \int_\Omega f(x)g(x)\, d\mu}$$

for all $f \in L^p(\Omega, \mu)$, where the integral is well-defined and finite. The norm of the functional f is given by

$$\|L\|_{(L^p)^*} = \|g\|_q.$$

Exercise

Let a, b real numbers such that $a < b$. The operator $T : L^2[a,b] \to \mathbb{R}$ is defined by

$$Tf = \int_a^b f(x)dx, \qquad f \in L^2[a,b].$$

a) Show that T si bonded. Compute $\|T\|$.

b) (do at home) According to the Riesz's Theorem, there exists a function $g \in L^2[a,b]$ such that

$$\boxed{Tf = \langle f, g \rangle,} \qquad \forall f \in L^2[a,b].$$

Find such a function g and verify that $\|g\|_{L^2} = \|T\|$.

L^2

Try by your self!

Solution

Point a). By Hölder's inequality we have

$$TF = \int_a^b f(x)\,dx$$

$$TF : L^2 \longrightarrow \mathbb{R}$$

$$\begin{aligned}
\|Tf\| &= \left|\int_a^b f(x)dx\right| \\
&\leq \int_a^b |\overset{g}{1} \cdot f(x)|dx \\
&\leq \left(\int_a^b 1^2\right)^{1/2} \underbrace{\left(\int_a^b |f(x)|^2 dx\right)}^{1/2} \\
&= \underbrace{\sqrt{b-a}}\|f\|_{L^2}.
\end{aligned}$$

$$\|Fg\|_{L^1} \leq \|f\|_{L^2} \|g\|_{L^2}$$

Hence $\boxed{T \text{ is bounded}}$. From the above inequality we get

$$\boxed{\|T\|_{L^2} \leq \sqrt{b-a}.} \tag{1}$$

Now we need to show that $\boxed{\|T\|_{L^2} \geq \sqrt{b-a}}$

Now, we consider the function $\boxed{h(x) = \frac{1}{\sqrt{b-a}}}$, $x \in (a, b)$. It is clear that $\boxed{h \in L^2(a, b)}$ and

well-defined

$$\|h\|_{L^2} = \left(\int_a^b (b-a)^{-1/2 \cdot 2}\right)^{1/2}$$

$$\|h\|_{L^2} = \left(\int_a^b \underbrace{(b-a)^{-1}}_{\text{constant}}\right)^{1/2} = 1,$$

$$(b-a)^{-1}(b-a)$$

and

$$\sqrt{b-a} \le \|T\|_{L^2}$$

$$|Th| = \left|\int_a^b \frac{1}{\sqrt{b-a}}\right| = \sqrt{b-a}.$$

$$\frac{b-a}{\sqrt{b-a}}$$

Hence

$$\boxed{\sqrt{b-a}} = |Th| \le \|T\|_{L^2} \underbrace{\|h\|_{L^2}}_{=1} = \|T\|_{L^2}. \tag{2}$$

Therefore by (1) and (2) we have

$$\boxed{\|T\|_{L^2} = \sqrt{b-a}.}$$

The following result is crucial:

Theorem: Let X and Y be two normed vector spaces, and let $T : X \to Y$ be a linear operator. The following conditions are equivalent:

a) The operator T is continuos in 0

b) The operator T is continuous at every point.

c) The following condition of boundedness is valid:

$$\boxed{\sup_{x \in X, x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X} < \infty.}$$

Exercise

Let us consider the following integral operator

$$Tf(x) = \int_\Omega k(x,y) f(y)\, dy, \tag{3}$$

where $\Omega \subseteq \mathbb{R}$ and $k : \Omega \times \Omega \to \mathbb{R}$ is a generic fixed function. This function is usually called kernel-function.

If

$$\boxed{\sup_{y\in\Omega} \int_\Omega |k(x,y)|\, dx < \infty.}$$

Then

$$\boxed{T : L^1(\Omega) \to L^1(\Omega).}$$

the operator T is linear and continuous

$$\Updownarrow$$

$$\|Tf\|_{L^1} \leq C \|f\|_{L^1}$$

Solution

By the linearity of the integral is obvious that the integral in (5) is linear. Now, we show the property of continuity of the operator T under suitable hypothesis on the kernel k.

$$
\begin{aligned}
\|Tf\|_{L^1(\Omega)} &= \int_\Omega |Tf(x)|dx \\
&= \int_\Omega \left| \int_\Omega k(x,y)f(y)dy \right| dx \\
&\leq \int_\Omega \int_\Omega |k(x,y)||f(y)|dydx. \qquad (4)
\end{aligned}
$$

definition

$$= \int_{\Omega} \int_{\Omega} |f(y)||k(x,y)|dx\, dy$$

Since the integrand function is positive by the Fubini-Tonelli we can exchange the integrals in (4) and we get

$$
\begin{aligned}
\|Tf\|_{L^1(\Omega)} &\leq \int_\Omega |f(y)| \left(\int_\Omega |k(x,y)| dx \right) dy \\
&\leq \int_\Omega |f(y)| \left(\sup_{y\in\Omega} \int_\Omega |k(x,z)| dx \right) dy \\
&= \left(\sup_{y\in\Omega} \int_\Omega |k(x,y)| dx \right) \int_\Omega |f(y)| dy \\
&= K \|f\|_{L^1(\Omega)},
\end{aligned}
$$

by hypothesis

where by hypothesis we have

$$
K = \sup_{y\in\Omega} \int_\Omega |k(x,y)| dx < \infty.
$$

Question

Exercise

Let us consider the following integral operator

$$Tf(x) = \int_{\Omega} k(x,y) f(y) dy, \tag{5}$$

where $\Omega \subseteq \mathbb{R}$ and $k : \Omega \times \Omega \to \mathbb{R}$ is a generic fixed function. If

$$\boxed{\int_{\Omega \times \Omega} |k(x,y)|^2 dxdy < \infty.}$$

Why?

Then

$$T : L^{\cancel{2}\,p}(\Omega) \to L^{\cancel{2}\,p}(\Omega).$$

the operator T is linear and continuous.

$T : L^p(\Omega) \to L^p(\Omega)$

By using the Hölder's inequality we get

$$|Tf(x)| = \left| \int_\Omega k(x,y) f(y) dy \right| \leq \|f\|_{L^2(\Omega)} \left(\int_\Omega |k(x,y)|^2 dy \right)^{1/2}.$$

Hence we get

$$\begin{aligned}
\|Tf\|_{L^2(\Omega)} &= \left(\int_\Omega |Tf(x)|^2 dx \right)^{1/2} \\
&\leq \left(\int_\Omega \|f\|^2_{L^2(\Omega)} \left(\int_\Omega |k(x,y)|^2 dy \right) dx \right)^{1/2} \\
&= \|f\|_{L^2(\Omega)} \left(\int_{\Omega \times \Omega} |k(x,y)|^2 dx dy \right)^{1/2} \\
&= K_1 \|f\|_{L^2(\Omega)},
\end{aligned}$$

$\int_\Omega \int_\Omega$

where by hypothesis we have

$$K_1 = \left(\int_{\Omega \times \Omega} |k(x,y)|^2 dx dy \right)^{1/2} < \infty.$$